Qualifying Exam in Microeconomics
Arizona State University
August 2018

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. Two firms or players indexed by $i = 1, 2$ may engage in a research and development (R&D) race. Each firm privately observes its type, strong (s) or weak (w). Types are independent and equally likely for both firms. The two players simultaneously and independently decide whether to participate in the R&D race. If at least one player chooses *not* to participate in the race, then no R&D takes place. (For a race to take place, both players must participate.) If both players participate in the race, the winner obtains a new patent and captures the market. (Firm i gets a patent only if both firms decide to enter the race, and player i wins.)

A strong firm always wins a race against a weak firm. If both firms are of the same type, each firm wins with equal probability.

Payoffs are as follows. To enter the R&D race, a firm must pay 2. This payment is only made when there is a race, that is to say when both firms choose to participate. A firm gets zero if it does not win and 10 if it wins.

(a) Describe the set of pure strategies available to a firm.

(b) Is $(N, Y), (N, Y)$ (that is both players use the same strategy) a Bayesian Nash equilibrium (BNE)?

(c) Find all pure strategy BNE.

(d) Is there a BNE of those identified that is less likely to be played than the others?

2. A decision maker chooses $x \geq 0$ to maximize

$$f(x,t) = tg(x) + h(x),$$

where $t \geq 0$ is a parameter, h is a real-valued function on $\mathbb{R}_+$, and g is a real-valued and *strictly increasing* function on $\mathbb{R}_+$.

(a) For this part, specialize to $g(x) = x$ and $h(x) = -\frac{1}{2}x^2 - x$.

  i. Solve for the optimal choice correspondence, *justifying your steps carefully.*

  ii. Describe a standard economic problem with this mathematical form. (This involves giving an economic interpretation for the variables t and x and the given function h.)

Return to the general case with the only assumptions on g and h being that g is strictly increasing.

(b) Does it follow that f satisfies the *strict single crossing property* (SSCP) in x and t? If so prove it. If not, then give additional assumptions on g and h under which f does satisfy the SSCP and prove your assertion.

(c) If f *does* satisfy the SSCP, does it follow that the solution correspondence will be increasing in t in this sense: for $t_1 > t_0 \geq 0$, if x_i maximizes $f(x, t_i)$ on $\mathbb{R}_+$, then $x_1 \geq x_0$? Explain in a sentence or two.

(d) Assume in addition that there is a solution to the maximization problem for every $t \geq 0$. Define

$$V(t) = \max_{x \geq 0} f(x,t),$$

the value function for the decision maker's optimization problem. What curvature properties (concavity, convexity, quasiconcavity, quasiconvexity), if any, does the value function V have? If it has one of these properties, prove it; otherwise give additional conditions on the functions g and h such that V does satisfy one of these properties and prove your assertion.

3. Let z be the aggregate excess demand function of an exchange economy with L goods when endowments are given by the vector ω.

Normalize prices so that $p_L = 1$ (rather than using the simplex) and let $\hat{p} = (p_1, \ldots, p_{L-1}) \in \mathbb{R}^{L-1}_{++}$. Define $\hat{z}(\hat{p}) = (z_1(\hat{p}, 1), \ldots, z_{L-1}(\hat{p}, 1))$.

Define $F : \mathbb{R}^{L-1}_{++} \times \mathbb{R}^{LI} \to \mathbb{R}^{L-1}$ by $F(\hat{p}, \omega) = \hat{z}(\hat{p})$ when endowments are given by the vector ω.

Define the equilibrium price correspondence by

$$Q(\omega) = \{\hat{p} \in \mathbb{R}^{L-1}_{++} : F(\hat{p}, \omega) = 0\}.$$

(a) Define what it means for the correspondence Q to have a closed graph.

(b) Prove that if F is continuous, Q has a closed graph.

4. An exchange economy has two consumers, 1 and 2, a *single* physical good, and two states of the world, 1 and 2. For $s = 1, 2$, the aggregate endowment of the good in state s is ω_s. Consumer 1's endowment is $(\omega_1, 0)$ and consumer 2's endowment is $(0, \omega_2)$. *In addition*, $\omega_1 = \omega_2 > 0$; in words, *there is no risk in the aggregate.*

The consumers each have subjective expected utility preferences on the consumption set $X = \mathbb{R}^2_+$ with the *same* belief $\pi = (\pi_1, \pi_2) >> 0$ about the likelihood of the states. Consumer i's von Neumann - Morgenstern utility is u_i. Each vN-M utility is differentiable with $u_i' > 0$ and $u_i'' < 0$, so the consumers are *strictly risk averse.*

Normalize the price vector $p = (p_1, p_2)$ so that its coordinates sum to 1.

(a) Define formally the notion of a competitive equilibrium for *this* economy. Draw an Edgeworth box for this economy and indicate the endowment.

(b) (i) Prove that $p = \pi$ is a competitive equilibrium price, and identify the associated allocation. (It turns out that the competitive equilibrium is unique, but we do not ask you to prove that fact.) (ii) Illustrate the competitive equilibrium in an Edgeworth box. (iii) Write down an expression for the expected utility of consumer i at the competitive equilibrium allocation.

(c) Suppose now that, *before* markets open, the consumers learn whether the state is 1 or the state is 2 (and update their beliefs accordingly). Prove that, if the consumers learn that the state is 1, then the unique competitive equilibrium price is $p = (1, 0)$. By symmetry the unique competitive equilibrium price if the consumers learn that the state is 2 is $p = (0, 1)$; you do not need to prove this second fact, but you should use it.

(d) Write down an expression for consumer i's *ex ante* expected utility from the situation in part (c); that is, expected utility before knowing what the state is, but knowing that once the state is revealed, the allocation will a competitive equilibrium allocation conditional on the state, as in (c).

(e) Which do the consumers prefer, to trade before the state is realized, as in part (b), or after the state is realized, as in part (c)? Explain.